ChatGPT said:

Assignment 003A — Your first binary file

We're going to write a few bytes directly to a file using C. This is our first step toward producing a playable WAV file.

For this assignment, **you write the code**. I'll give you the specifications and the C functions you'll need.

Objective

Create a file named `test.bin` containing exactly four bytes:

```
Plain text

Hex:       41 42 43 44
Decimal:   65 66 67 68
ASCII:      A  B  C  D
```

Your task

Create a function with this signature:

```c
int write_test_file(const char *filename);
```

The function should:

1. Open a file for binary writing.
2. Check whether opening the file succeeded.
3. Write four bytes containing `0x41`, `0x42`, `0x43`, `0x44`.
4. Close the file.
5. Return `0` for success, nonzero for failure.

Call it from `main()` and create `test.bin`.

C functions you'll need

Opening the file:

C

```c
FILE *file = fopen(filename, "wb");
```

`fopen()` returns a pointer to a `FILE` object, or `NULL` if it fails.

Writing the bytes:

C

```c
fwrite(ptr, size, count, file);
```

The arguments are:

Argument	Meaning
`ptr`	Address of the data to write
`size`	Size of each element in bytes
`count`	Number of elements
`file`	File pointer returned by `fopen()`

`fwrite()` returns the number of complete elements successfully written.

Closing the file:

C

```c
fclose(file);
```

Use `unsigned char` for your four bytes. Each `unsigned char` occupies exactly one C byte.

Verify your work

Once it compiles, run:

Bash

```bash
./subosc
xxd -g 1 test.bin
```

You should see something resembling:

```
00000000: 41 42 43 44  ABCD
```

The first column is the byte offset. The middle shows the hexadecimal bytes, and the right side shows their printable ASCII interpretation.

Also try:

```bash
wc -c test.bin
```

It should report `4` bytes.

Two things to figure out yourself

I deliberately haven't supplied the full function.

1. How will you create an array containing those four bytes and pass its address to `fwrite()`?
2. How will you use the return value from `fwrite()` to determine whether all four bytes were written?

You already have the C knowledge needed for both.

Stopping point: Show me your `write_test_file()` function and the output of `xxd`.

After that, we'll examine a WAV header and begin turning your oscillator into something you can hear.